

Introduction to Aerial Robotics

Lecture 1

Shaojie Shen
Associate Professor
Dept. of ECE, HKUST



3 February 2026

Logistics

About this Course...

- This course is about:
 - Autonomy for aerial robots
 - Modeling and control of multi-rotor aerial robots
 - Path and trajectory planning for multi-rotor aerial robots
 - Algorithms for vision-based state estimation
 - Algorithms for multi-sensor fusion
 - Real-time software implementations for autonomous aerial robots
 - Lots of fun with robots 😊
- This course is NOT about:
 - Aerodynamics 😞
 - Aircraft design 😞
 - Electromechanical systems of aerial robots 😞

Course structure

- Weekly lectures + lots of projects + a midterm exam
- Grading Scheme:
 - Midterm exam: 20%
 - Project 1: 30%
 - Project 2: 20%
 - Project 3: 30%
- More details can be found on the Canvas homepage.

Teaching Team

- **Instructor:**

- Shaojie Shen (eeshaojie@ust.hk)

- Office Hour: by appointment



- **Teaching Assistants:**

- Yang Xu (yxuew@connect.ust.hk)

- Office Hour: by appointment

- Pusen Gao (pgaoak@connect.ust.hk)

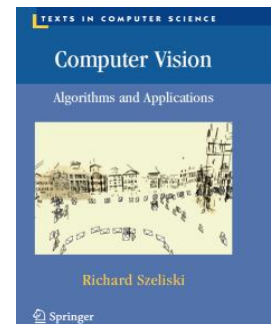
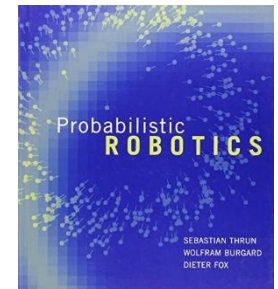
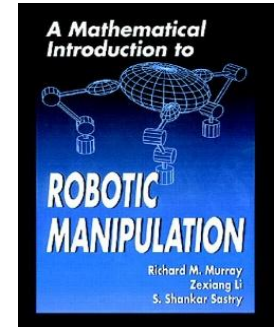
- Office Hour: by appointment

Administrative Stuff

- **Lecture:**
 - Tuesday 1:30 pm - 4:20 pm
 - Rm 5566, Lift 27-28 (30)
 - Students who are not registered are welcome to sit in the lectures, but their assignments will not be graded
- **Labs:**
 - Wednesday 06:00 pm - 08:50 pm
 - Robotics Institute Flight Area
 - We will not have lab sessions every week, refer to Canvas site for details
- **Course Website:**
 - <https://canvas.ust.hk/courses/68555>

(Non-Compulsory) Reference Books

- A Mathematical Introduction to Robotic Manipulation
 - Richard M. Murray, Zexiang Li, S. Shankar Sastry;
- Probabilistic Robotics;
 - Sebastian Thrun, Wolfram Burgard, Dieter Fox;
- Computer Vision: Algorithms and Applications
 - Richard Szeliski
 - <http://szeliski.org/Book/>



Workload & Expectation

- **Expected Student Background:**

- Linear algebra
- Probability
- Python programming skills
- C++ programming skills (VERY IMPORTANT)
- Linux
- Love robots 😊

- **Workload:**

- Attend lectures
- Lots of project work
- Have fun with robots 😊

Programming!

- Python
- C++
- Robot Operating System (ROS)

ROS

Requires coding on both desktop PC and
embedded platforms



Overview of Aerial Robotics

Aerial Robots = Military Drones?

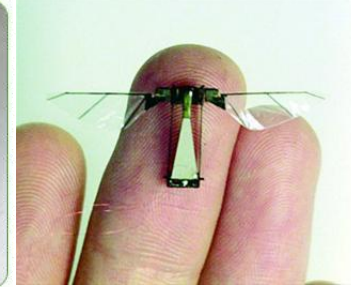
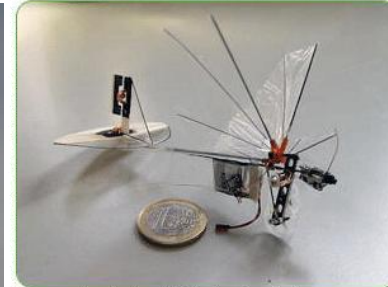
- Mostly Big
- 10+ Hours of Flight Time
- Remote Control
 - Waypoint planning
 - Some joystick
 - Flight crew 4-10



“Drones mischaracterize what these things are. They're not dumb. Nor are they unmanned, actually. They're remotely piloted aircraft.”

-- Gen. Norton Schwarz, August 10, 2012

Unmanned Aerial Vehicles

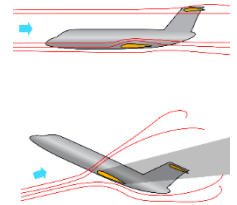


Unmanned Aerial Vehicles

- Types of UAVs

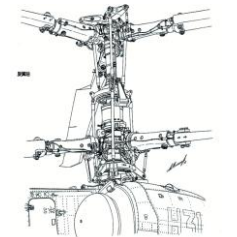
- Fixed wing

- Long flight time, large payload, self-stabilized system ☺
 - Requires runway, may stall, needs aerodynamic design ☹



- Helicopter

- Vertical takeoff and landing ☺
 - OK flight time and payload
 - Complex mechanical system, unstable systems ☹



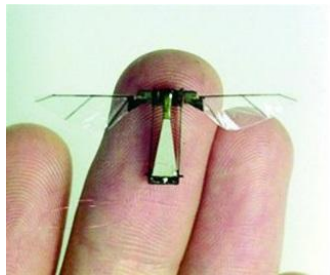
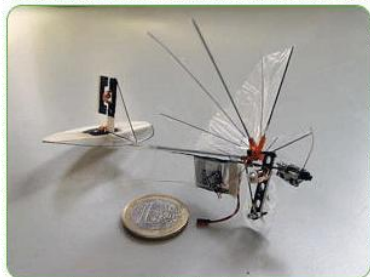
- Multi-rotor

- VTOL, simple mechanical design ☺
 - Hard to scale up
 - Short flight time, small payload, unstable system ☹



Unmanned Aerial Vehicles

- Anything even better?
 - Vertical Take-Off and Landing (VTOL) UAV
 - VTOL + long flight time + long range 😊
 - Technology not well developed ☹️
 - Flapping Wing / Bio-Inspired UAVs
 - Suitable for small platforms 😊
 - Technology not well developed ☹️

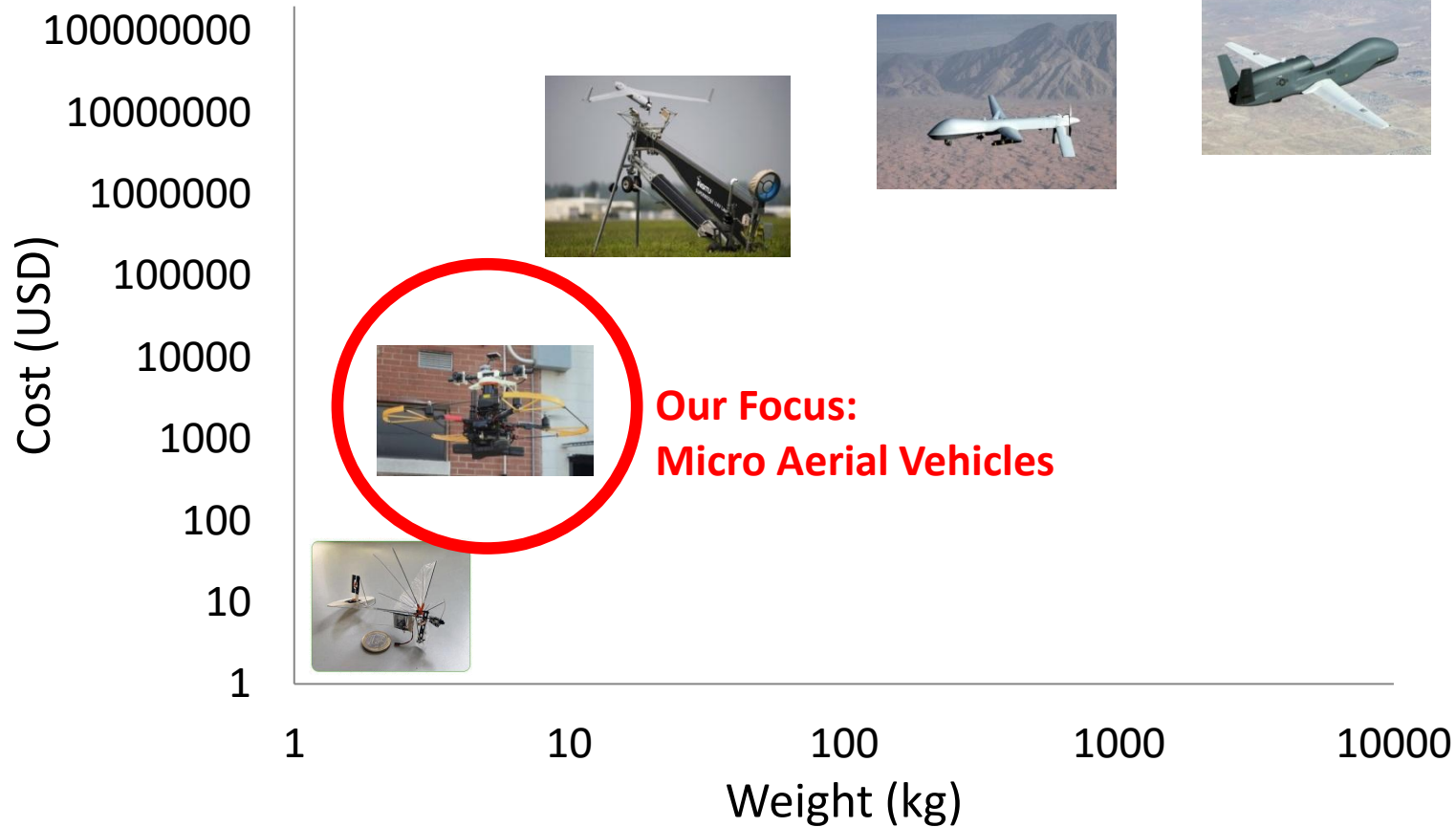


Contents [\[hide\]](#)

- 1 Crashes
 - 1.1 June 1991
 - 1.2 July 1992
 - 1.3 April 2000
 - 1.4 December 2000
 - 1.5 April 2010
 - 1.6 April 2012
 - 1.7 June 2012
- 2 Other accidents and notable incidents
 - 2.1 March 2006
 - 2.2 July 2006
 - 2.3 March 2007
 - 2.4 November 2007
 - 2.5 2009
 - 2.6 October 2014
 - 2.7 May 2015
- 3 References
- 4 External links

Unmanned Aerial Vehicles

- Civilian market of \$100 billion USD
- Regulations are getting mature



Multi-Rotor Micro Aerial Vehicles

- Small size (<1m)
- Adequate payload (1-5kg)
- Low cost (< 10k USD)
- Safe
- Superior mobility



Inspection



Search and Rescue



Transportation



Aerial Photography



Law enforcement



Agriculture

Development of Multi-Rotors

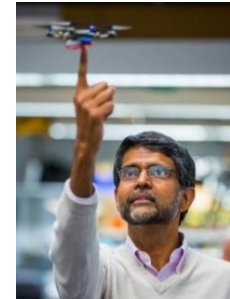
- de Bothezat helicopter (1922)
 - Flight altitude 5m during demonstration
 - Fully manual controlled unstable system ☹️
- Nothing happened in the next few decades
 - Almost impossible for manual control ☹️
 - Multi-rotors cannot be scaled up ☹️
 - No suitable sensors (tradition IMUs are big and expensive) ☹️
- STARMAC @ Stanford University (2004)
 - Thanks to the development of smart mobile devices
 - MEMS sensors + embedded computers



Development of Multi-Rotors

- The Role of Academia

- University of Pennsylvania
 - Vijay Kumar (Planning, control, swarm)
- Massachusetts Institute of Technology
 - Jonathan How (modelling, control)
 - Nicolas Roy (perception)
- University of California, Berkeley
 - Claire Tomlin (control, policies)
- ETH Zurich
 - Roland Siegwart (perception, control)
 - Raffaello D'Andrea (control, swarm)
 - Marc Pollefeys (perception, Pixhawk)
- Univ. of Zurich
 - Davide Scaramuzza (perception, control)
- Hong Kong University of Science Technology
 - Zexiang Li (founded DJI with Frank Wang)
 - I actually want to put myself here...
- Also involves other areas such as communication, material science, mechanical engineering, electronics, aerospace engineering, system engineering...



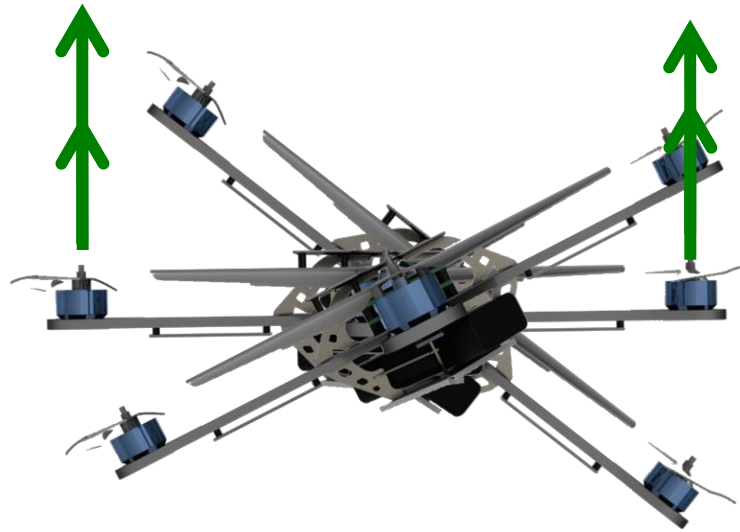
Commercialization of Aerial Robots



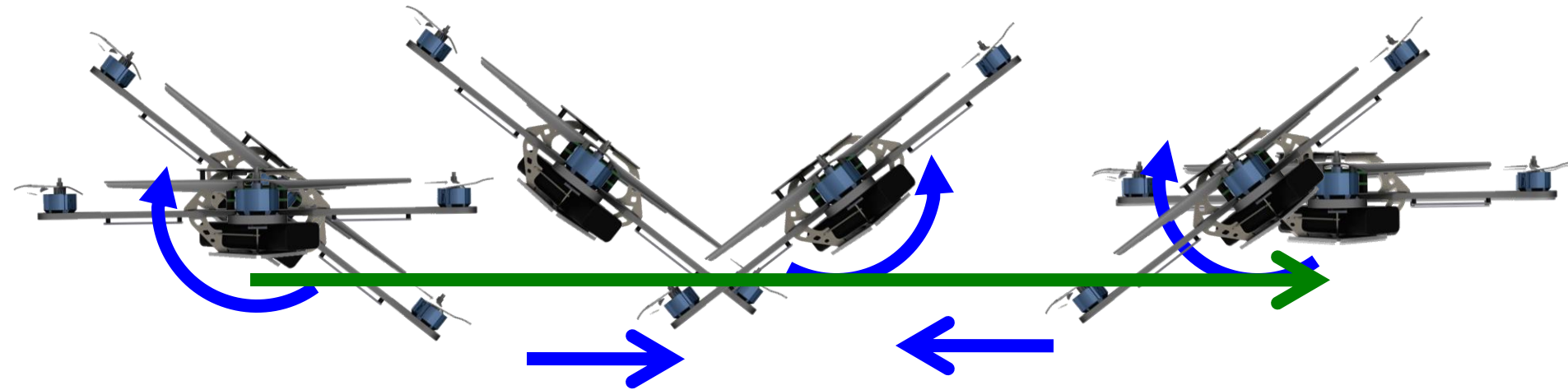
Quadrotors



Quadrotors - Rotation



Quadrotors - Translation



Towards Autonomous Flight...

- Sensing and perception

Part 2

- State estimation

Part 3

- Path planning

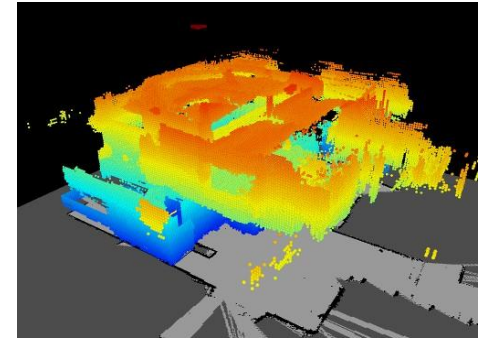
- Trajectory generation

Part 1

- Control

- System integration

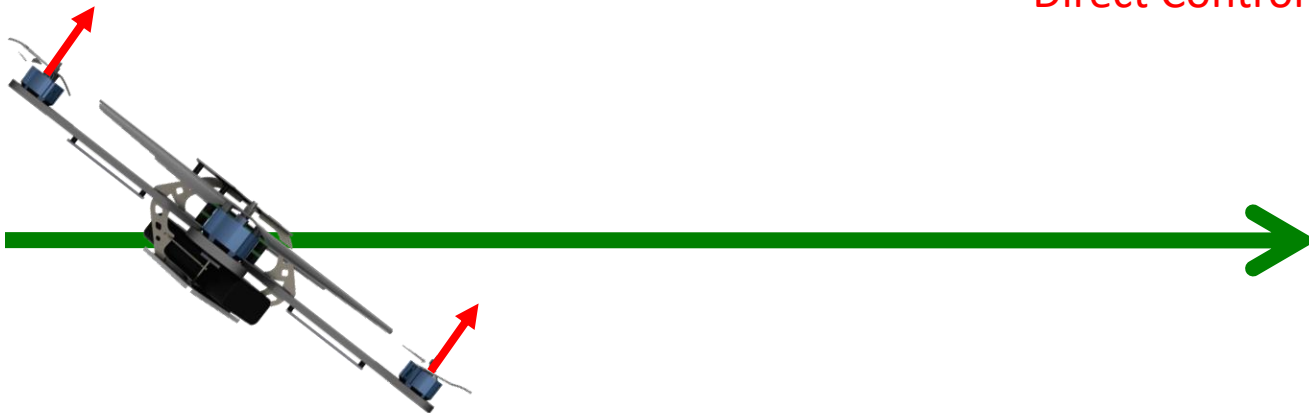
Projects!



How to fly? – Dynamics & Control

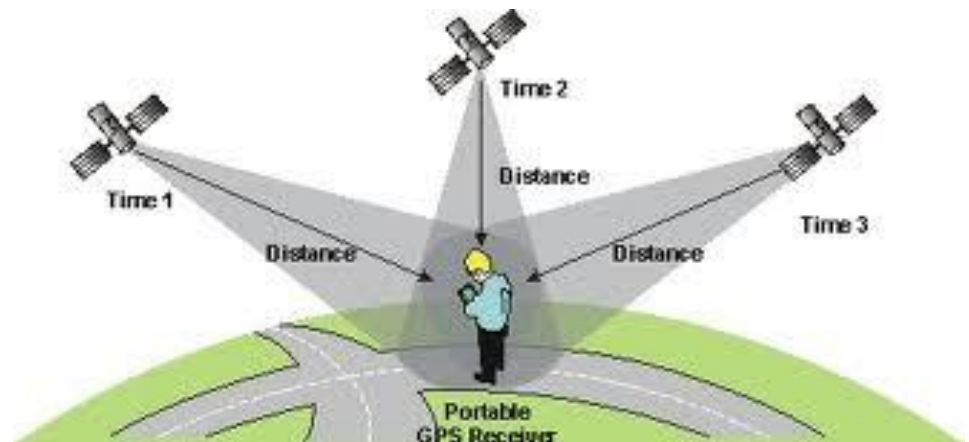
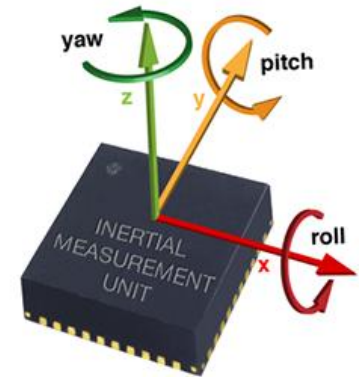
Derivative	Translation	Rotation	Thrust
0	Position → Control Target		
1	Velocity		
2	Acceleration	Rotation	
3	Jerk	Angular Velocity	
4	Snap	Angular Acceleration	Differential Thrust
5	Crackle	Angular Jerk	Change in Thrust

Direct Control Input

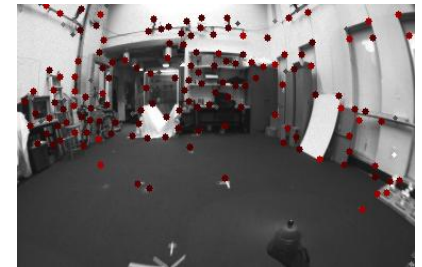
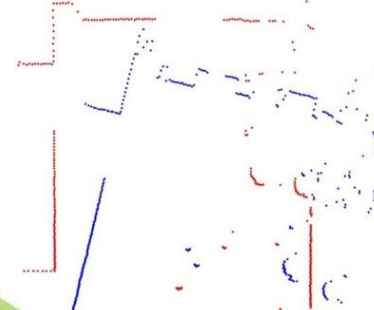
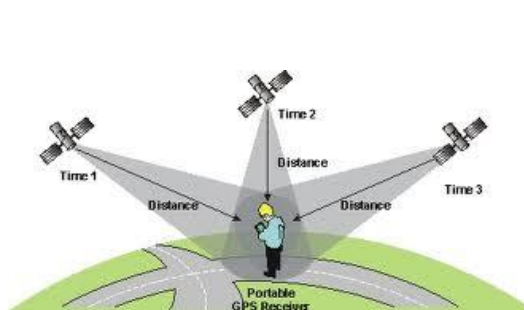
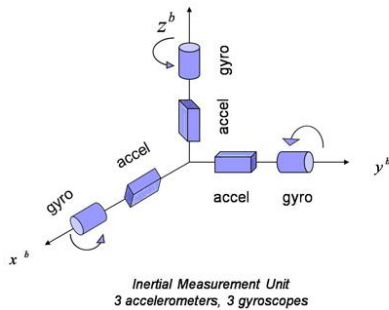


How to Fly? – Sensing & Estimation

- Answer the key question: **Where am I?**
- Proprioceptive sensors
 - Low cost Inertial Measurement Unit
- Exteroceptive sensors
 - GPS
 - Magnetometer
 - Barometer
 - Cameras
 - Laser range finders
 - etc.
- Processors
- Algorithms



How to Fly? – Sensing & Estimation



Inertial Measurement Unit

GPS

Laser Range Finder

Camera

How to Fly? – Navigation

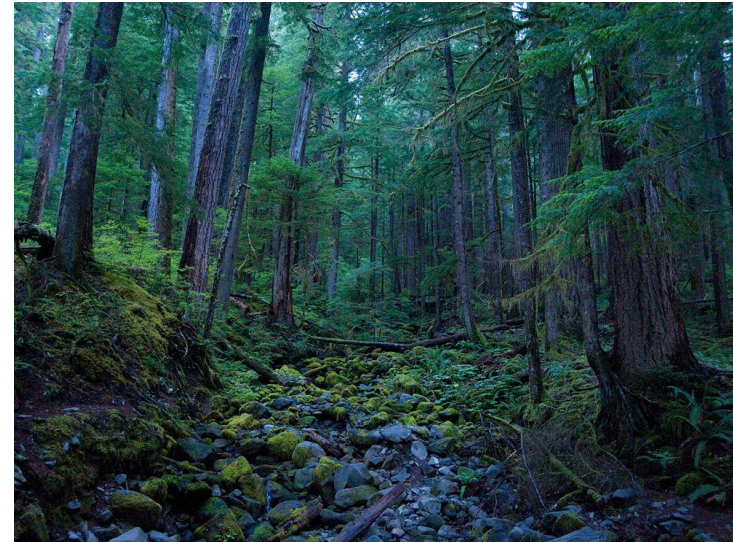
- Remote control
 - Requires line of sight
 - Requires communication link
 - Requires skilled pilots
- Inertial navigation
 - Requires aviation grade IMU
 - Heavy and expensive
- GPS-based navigation
 - Waypoint following
 - No obstacle avoidance
 - GPS can be unreliable



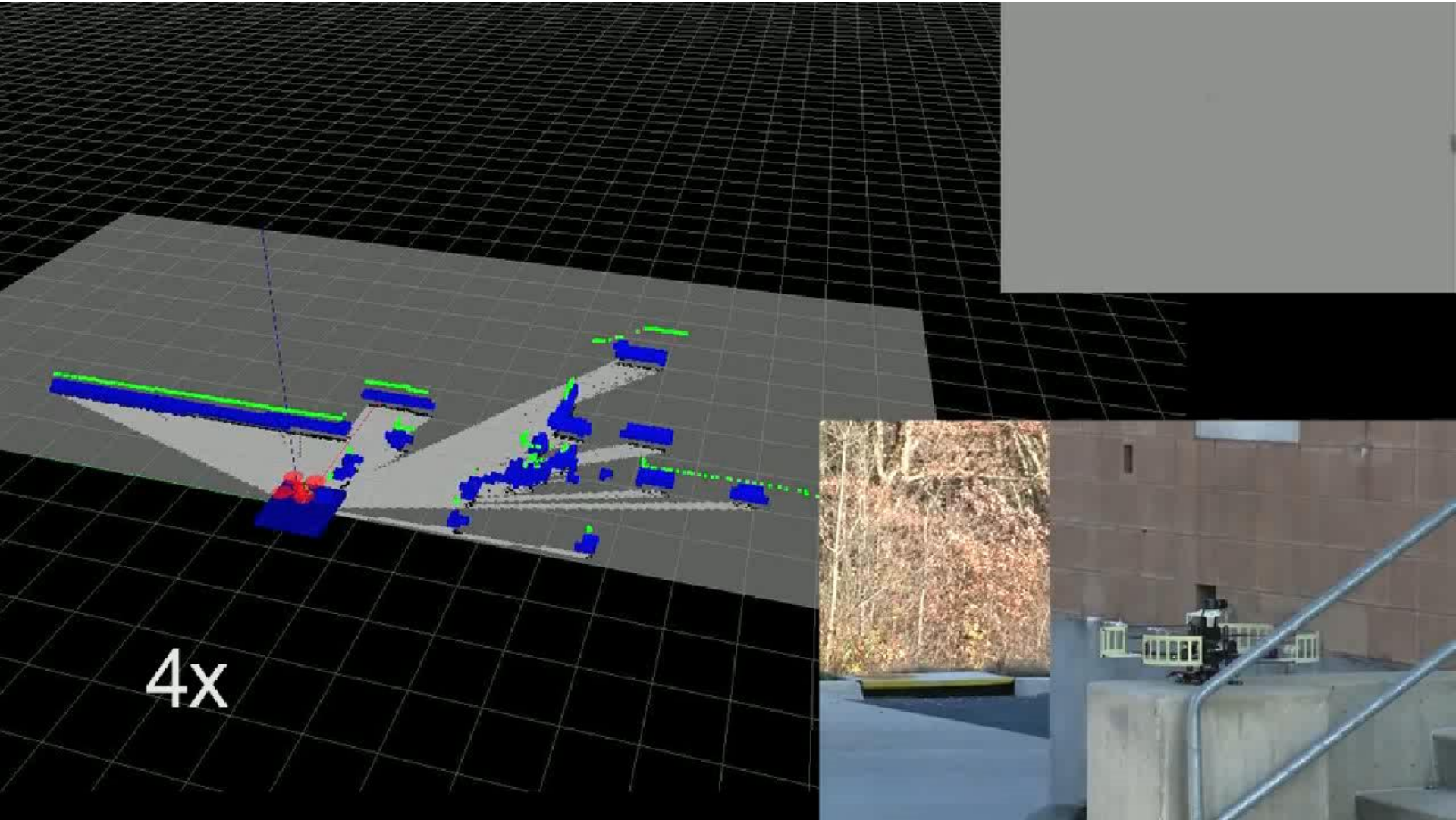
How to Fly? – GPS-based Navigation



How to Fly? – GPS-denied Navigation



How to Fly? – Laser-based Navigation



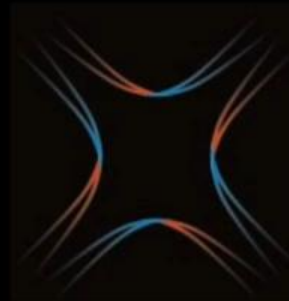
How to Fly? – Laser-based Navigation

Robust and Efficient Quadrotor Trajectory Generation for Fast Autonomous Flight

Boyu Zhou, Fei Gao, Luqi Wang, Chuhao Liu and Shaojie Shen



香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY



香港科技大學-
大疆創新科技聯合實驗室
HKUST-DJI JOINT
INNOVATION LABORATORY

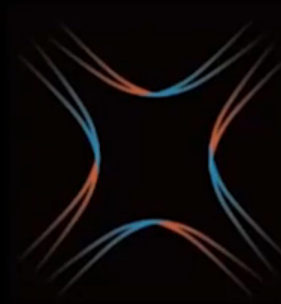
How to Fly? – Vision-based Navigation

RAPTOR: Robust and Perception-aware Trajectory Replanning for Quadrotor Fast Flight

Boyu Zhou, Jie Pan, Fei Gao and Shaojie Shen



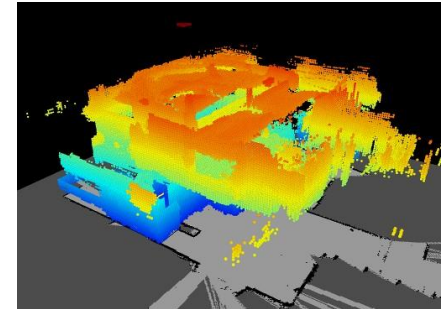
香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY



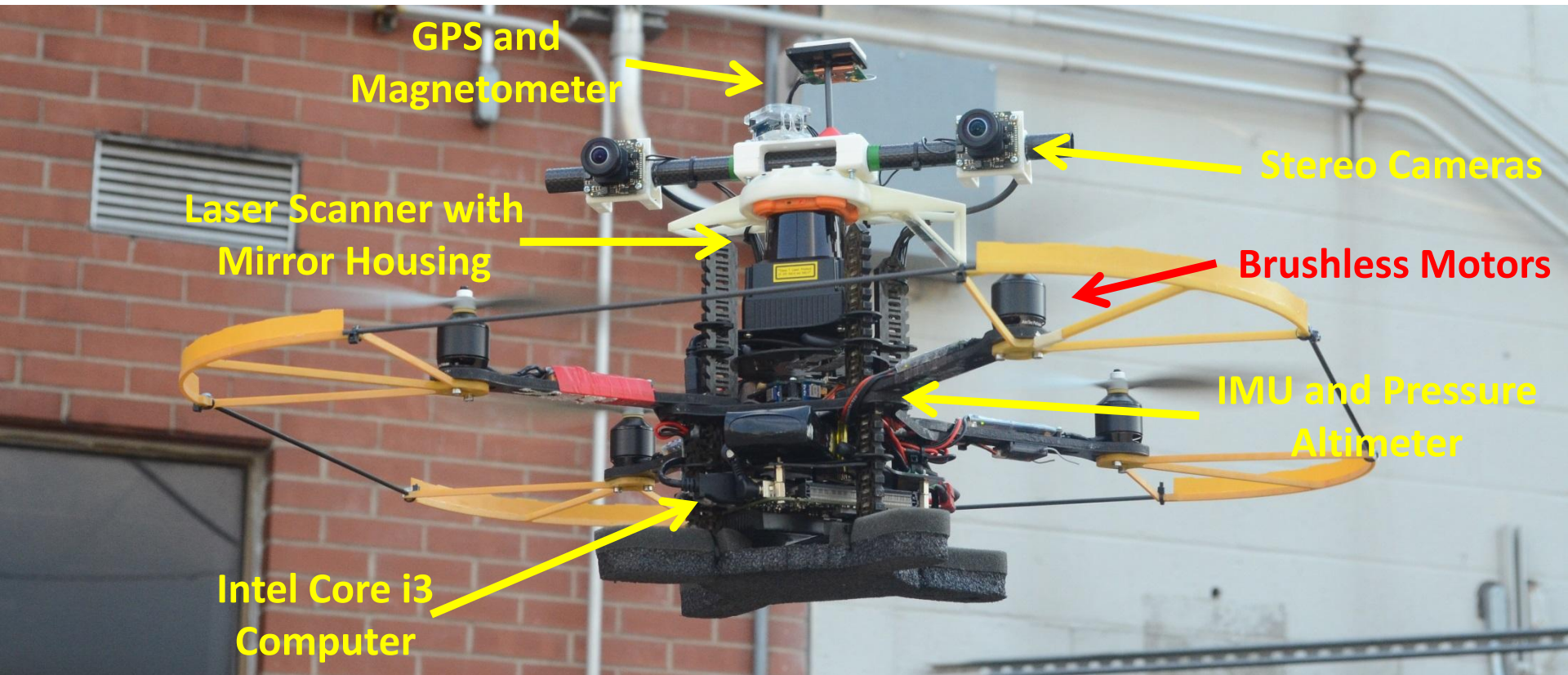
香港科技大學-
大疆創新科技聯合實驗室
HKUST-DJI JOINT
INNOVATION LABORATORY

Challenges

- Sensing & Perception
 - 3D sensing & mapping
- State Estimation & Localization
 - Low latency & high accuracy
- Obstacle avoidance
 - Complex & unknown environments
- Trajectory Control
 - Aggressive maneuvers
 - Smooth trajectory tracking
- System integration
 - Limited sensing & computation
 - Autonomous operations



A Flying Robot



Goal



“Prometheus”

Autonomous Exploration

FUEL: Fast UAV Exploration using Incremental Frontier Structure and Hierarchical Planning

Boyu Zhou, Yichen Zhang, Xinyi Chen and Shaojie Shen



香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

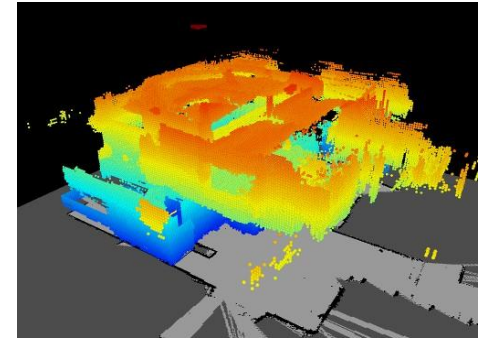


香港科技大學-
大疆創新科技聯合實驗室
HKUST-DJI JOINT
INNOVATION LABORATORY

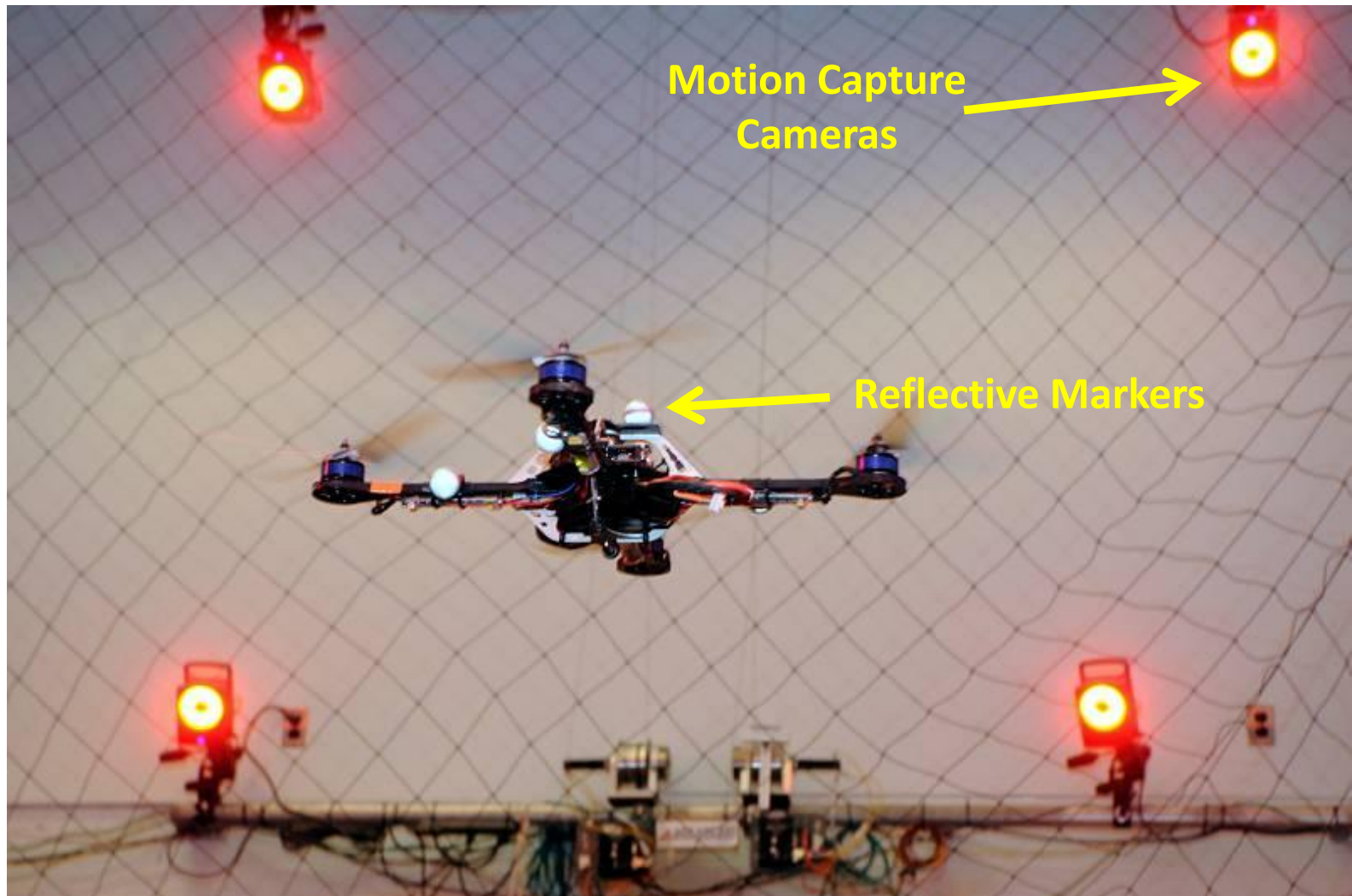
Autonomous Exploration

Towards Autonomous Flight...

- Sensing and perception
- State estimation
- Path planning
- Trajectory generation
- Control
- System integration



Bypass the Sensing Problem



Robust Control

Science Robotics

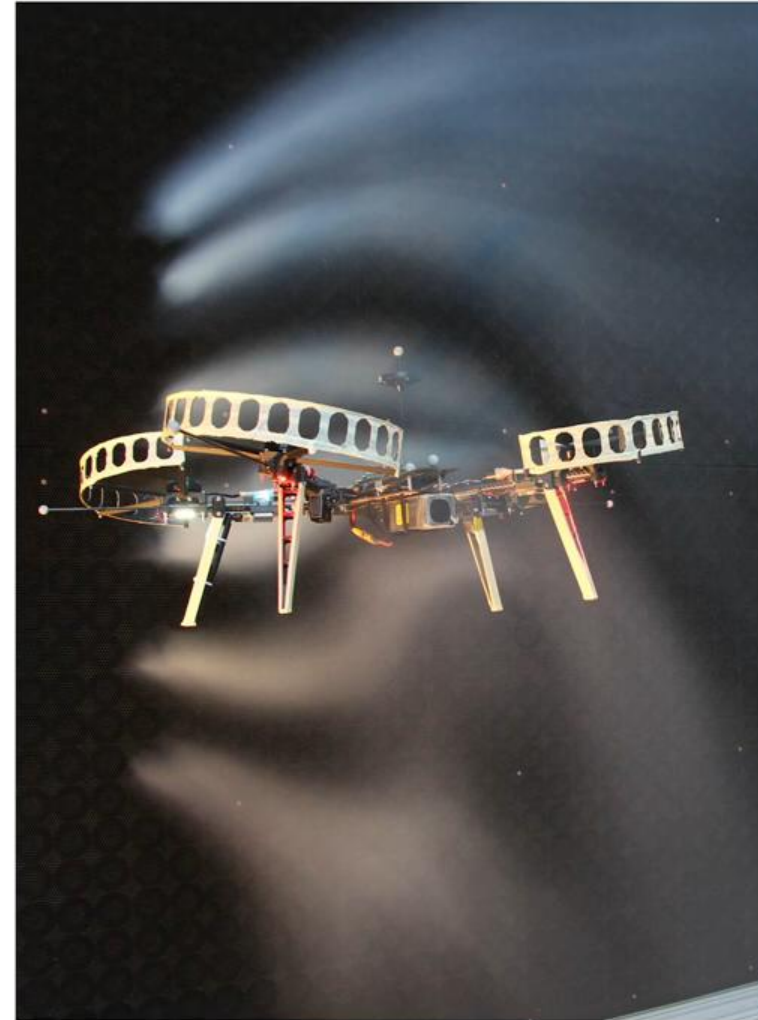
Caltech

Neural-Fly Enables Rapid Learning for Agile Flight in Strong Winds

Michael O'Connell*, Guanya Shi*, Xichen Shi,
Kamyar Azizzadenesheli, Anima Anandkumar,
Yisong Yue, Soon-Jo Chung†

*Equal contribution and alphabetical order

†Corresponding email: sjchung@caltech.edu



Robust Control for Drone Racing

Impact-aware Control

Impact-Aware Planning and Control for Aerial Robot With Suspended Payload

Haokun Wang^{1†}, Haojia Li^{1†}, Boyu Zhou^{2*}, Fei Gao^{3*}, Shaojie Shen¹



Ball Catching

Catch Planner: Catching High-Speed Targets in the Flight

Huan Yu*, Pengqin Wang*, Jin Wang†, Jialin Ji, Zhi Zheng, Jie Tu,
Guodong Lu, Jun Meng, Meixin Zhu, Shaojie Shen, and Fei Gao†



浙江大學 机器人研究院
ZHEJIANG UNIVERSITY ROBOTICS INSTITUTE



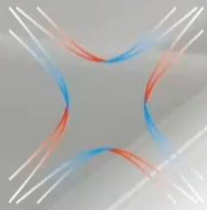
香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

FASTLAB

Autonomous Flights in Tunnels

Autonomous Flights inside Narrow Tunnels

Luqi Wang, Yan Ning, Hongming Chen, Peize Liu,
Yang Xu, Hao Xu, Ximin Lyu and Shaojie Shen



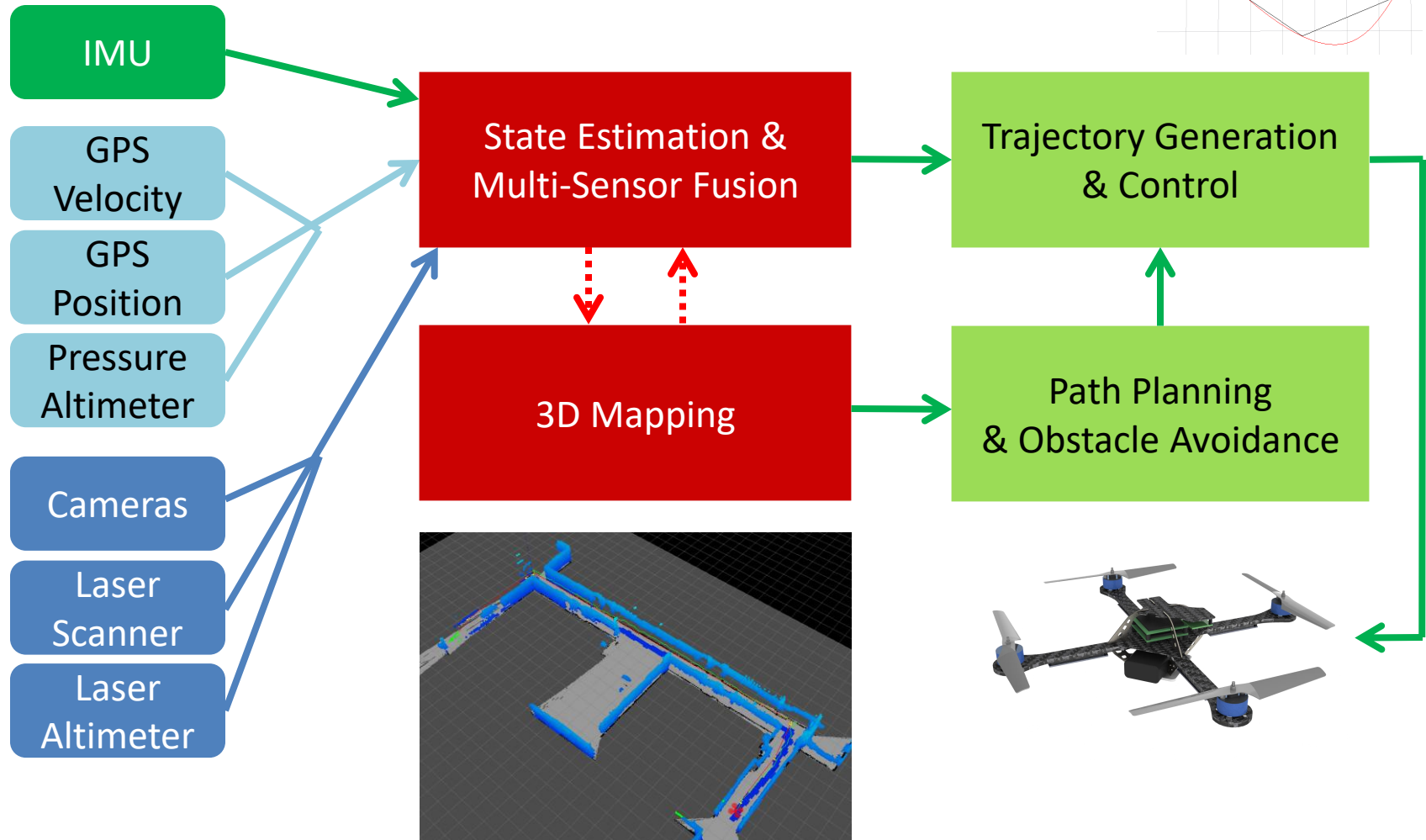
HKUST-DJI JOINT
INNOVATION LABORATORY
香港科技大學-大疆創新科技聯合實驗室



中山大學
SUN YAT-SEN UNIVERSITY



The Complete System



Search and Rescue

Autonomous Inspection

FC-Planner: A Skeleton-guided Planning Framework for Fast Aerial Coverage of Complex 3D Scenes

Chen Feng, Haojia Li, Jinqi Jiang, Xinyi Chen, Shaojie Shen, and Boyu Zhou



STAR



HKUST - DJI JOINT
INNOVATION LABORATORY
香港科技大學 - 大疆創新科技聯合實驗室



Autonomous Reconstruction

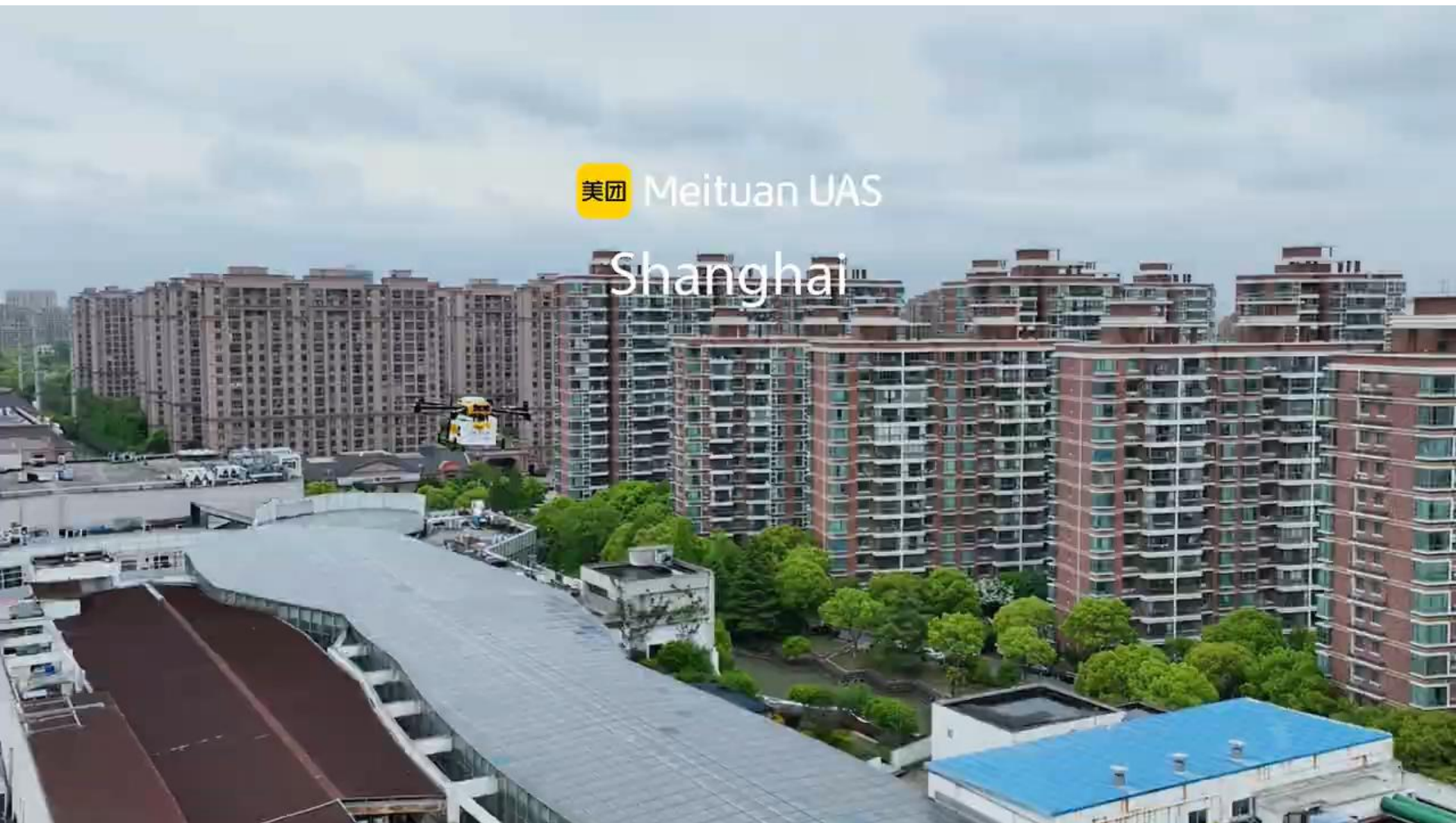


Feng C., et al. IJRR (In submission), 2026

Industrial Application

Precision Farming

Delivery

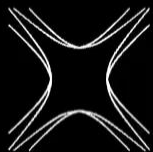


Meituan UAS Application

Payload System for Delivery

AD-Planner: Adaptive In-flight Delivery Using a Quadrotor with a Suspended Payload

Haojia Li, Yang Xu, Chen Feng, Boyu Zhou*, and Shaojie Shen



HKUST-DJI JOINT
INNOVATION LABORATORY



香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

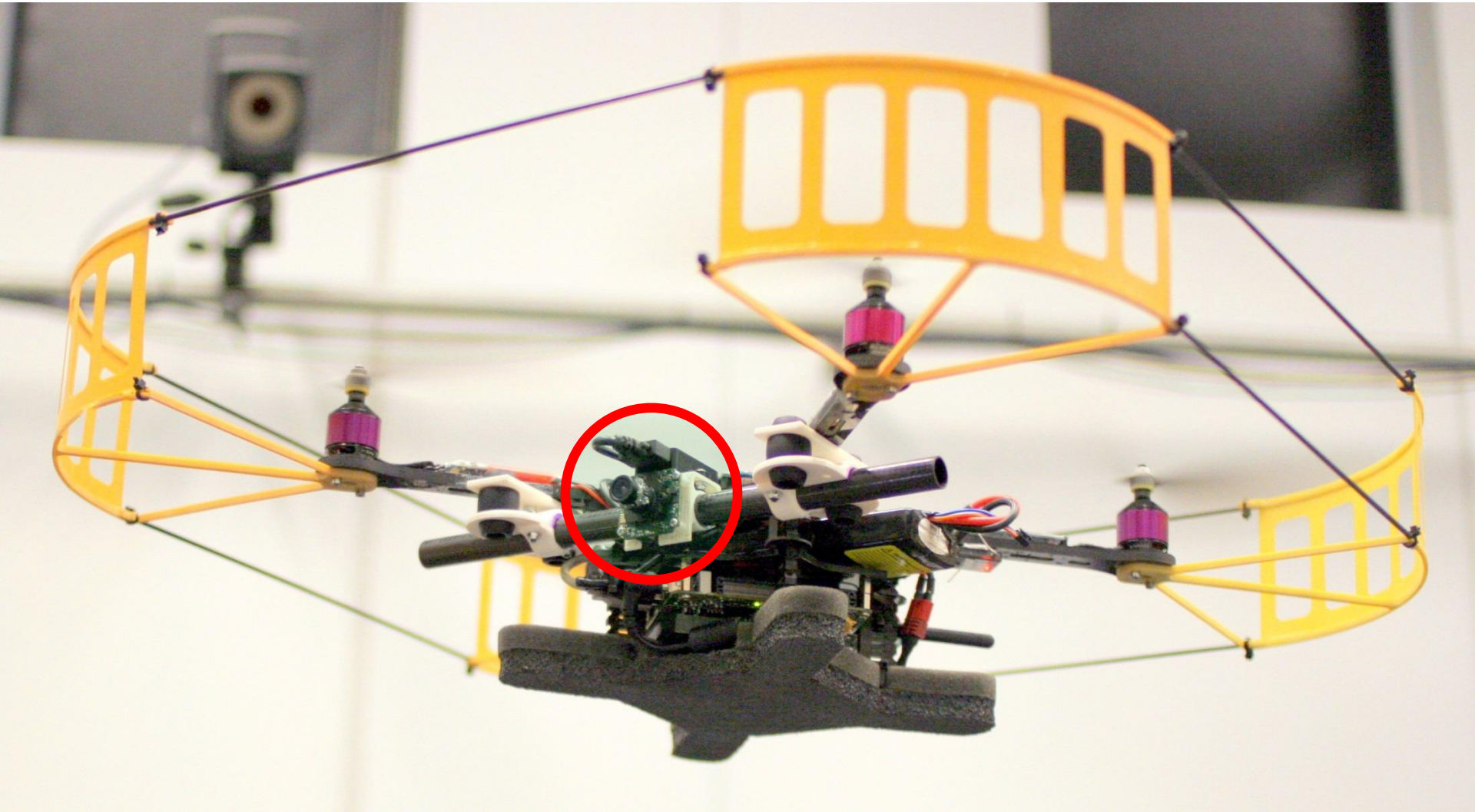
Small and Autonomous



Processing: Not quite there yet, but remember the **Moore's Law**!
Sensing: **YES!**



Minimum Sensing: 1 Camera + 1 IMU



Minimum Sensing: 1 Camera + 1 IMU

Autonomous Aerial Navigation Using Monocular Visual-Inertial Fusion

Yi Lin, Fei Gao, Tong Qin, Wenliang Gao,
Tianbo Liu, William Wu, Zhenfei Yang and Shaojie Shen



High resolution video available at:
<http://ece.ust.hk/~eeshaojie/jfr2017yi.mp4>

Minimum Omnidirectional Sensing

Dual-Fisheye Stereo + IMU

Dual-Fisheye Omnidirectional Stereo

Wenliang Gao and Shaojie Shen



香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

High resolution video available at
<http://www.ece.ust.hk/~eeshaojie/iros2017wenliang.mp4>

Minimum Omnidirectional Sensing

Dual-Fisheye Stereo + IMU

RC Control



Requires DJI Neo 2 Digital Transceiver, along with the designated remote controller and goggles. These can be obtained by purchasing specific controller or by buying the designated items separately.

Swarm



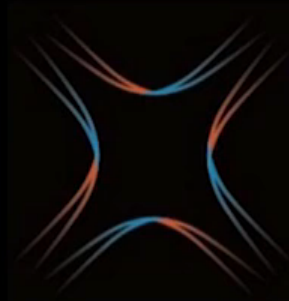
Swarm Localization

Omni-swarm: A Decentralized Omnidirectional Visual-Inertial-UWB State Estimation System for Aerial Swarms

Hao Xu, Yichen Zhang, Boyu Zhou, Luqi Wang,
Xinjie Yao, Guotao Meng, Shaojie Shen



香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY



香港科技大學-
大疆創新科技聯合實驗室
HKUST-DJI JOINT
INNOVATION LABORATORY

Swarm Navigation

Utopia: Firefly



Course Outline

Course Outline

- Dynamics, Planning & Control

- Vision

- Estimation

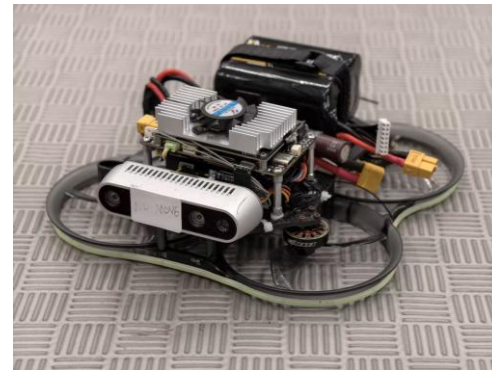
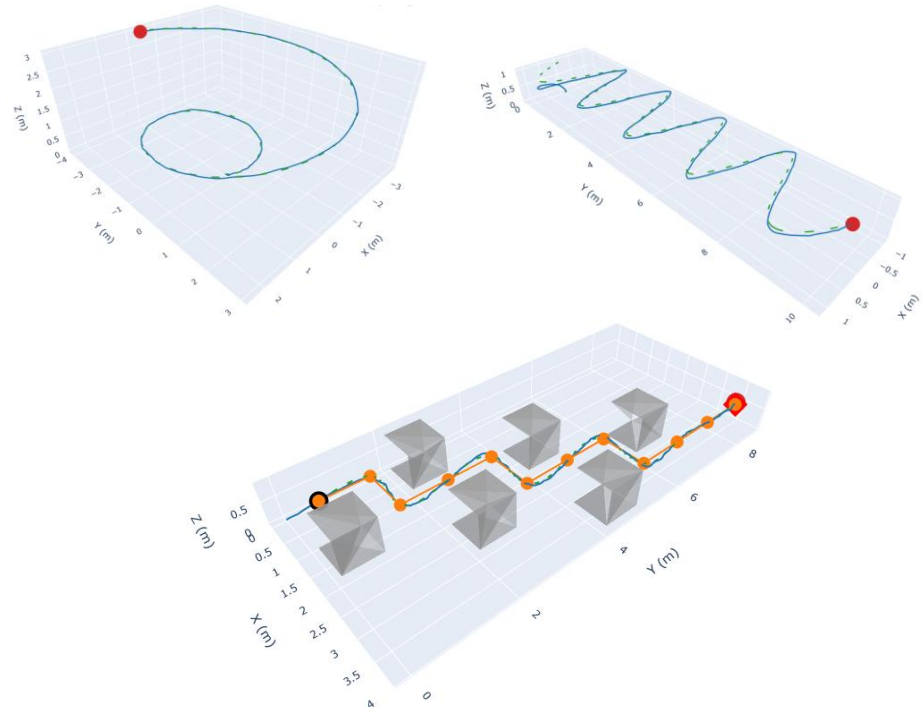
Week	Lecture Date Tue 13:30-16:20	Topic	Assignment (due 23:59 on Friday of the corresponding week)	Lab Wed 18:00-20:50
1	2/3	Introduction		No Lab
2	2/10	Rigid Body Transformation Quaternion Quadrotor Modeling		No Lab
3	2/17	Lunar New Year's Day (No class)		No Lab
4	2/24	Control Basics Quadrotor Control Trajectory Generation	Project 1 Phase 1 Out	No Lab
5	3/3	Trajectory Generation Path Planning	Project 1 Phase 1 Due Project 1 Phase 2 Out	No Lab
6	3/10	Camera Modeling & Calibration Feature Detection & Matching	Project 1 Phase 2 Due Project 1 Phase 3 Out	No Lab
7	3/17	Midterm	Project 1 Phase 3 Due Project 1 Phase 4 Out	Lab Tutorial 1: Robot Assembly
8	3/24	Multi-View Geometry Pose Estimation	Project 2 Phase 1 Out	Lab Tutorial 2: Prepare P1P4
9	3/31	Optical Flow Dense Stereo	Project 2 Phase 1 Due Project 2 Phase 2 Out	Free Lab Time
10	4/7	Midterm Break (No class)		No Lab
11	4/14	Probability Basics Bayesian Inferencing Kalman Filter	Project 2 Phase 2 Due Project 3 Phase 1 Out	Free Lab Time
12	4/21	Extended Kalman Filter Augmented State EKF Particle Filter	Project 1 Phase 4 Due Project 3 Phase 2 Out	Free Lab Time
13	4/28	SLAM	Project 3 Phase 3 Out Project 3 Phase 1 Due	Lab Tutorial 3: Prepare P3P3
14	5/5	x	Project 3 Phase 2 Due	Free Lab Time
15	5/12	x	Project 3 Phase 3 Due	Free Lab Time

Course Outline

- Dynamics, Control, and Planning
 - Understand 3D rigid body transformation
 - Derive the model of quadrotor aerial robots
 - Able to hover the quadrotor given perfect localization
 - Able to generate and track smooth trajectories given user-defined waypoints
 - Able to generate smooth and safe paths and trajectories given obstacle locations
 - Project 1
- Vision
 - Understand how cameras capture the world
 - Understand feature-based and optical flow-based image processing pipeline
 - Able to use only images to compute camera position, orientation, and velocity
 - Project 2
- Estimation
 - Understand how to incorporate probability principles into robotics
 - Able to integrate heterogeneous noisy measurements to get robust estimates of the platform motion
 - A light touch of simultaneous localization and mapping (SLAM)
 - Project 3

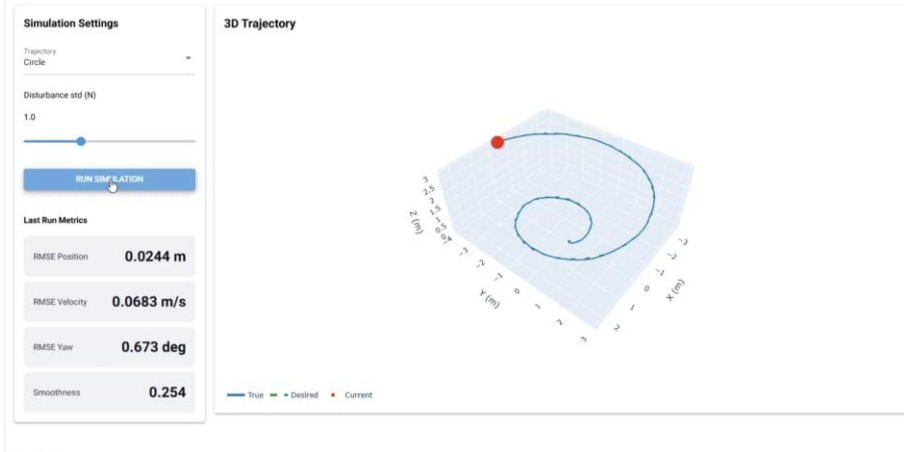
Project 1

- Phase 1: Hovering and trajectory tracking of a simulated quadrotor
 - Python, offline
- Phase 2: Trajectory generation and tracking of a simulated quadrotor
 - Python, offline
- Phase 3: Obstacle avoidance using A* path planning and smooth trajectory generation
 - Python, offline
- Phase 4: Fly the robot!

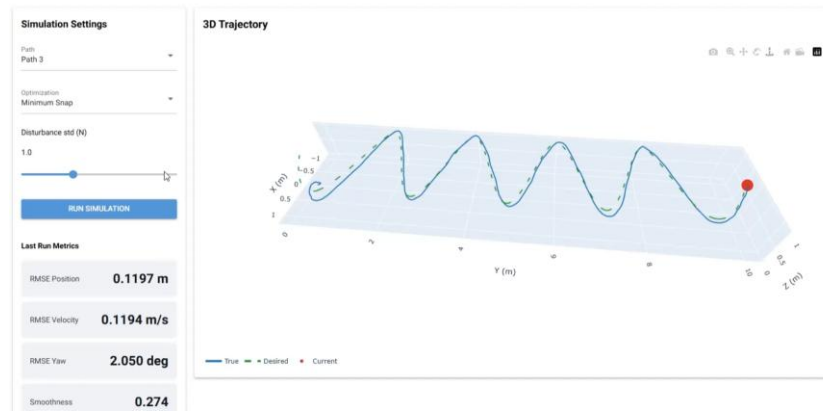


Project 1

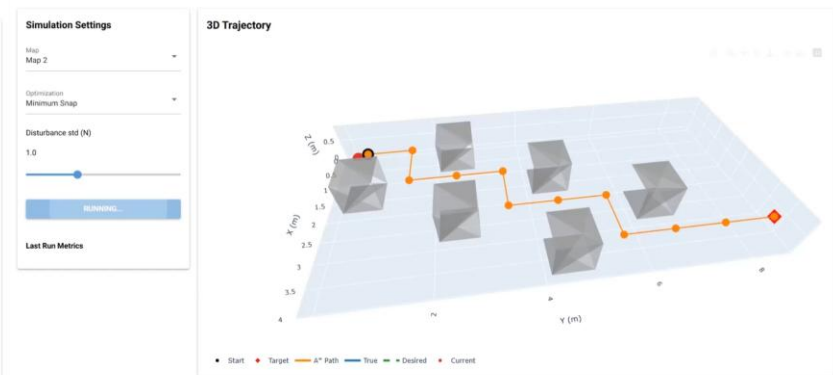
HKUST ELEC5660 – Quadrotor Simulator



HKUST ELEC5660 – Quadrotor Simulator



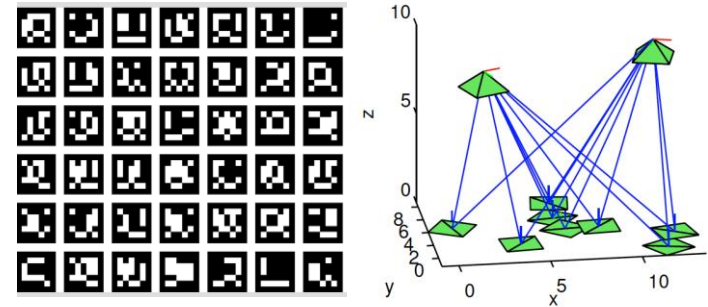
HKUST ELEC5660 – Quadrotor Simulator



Analysis

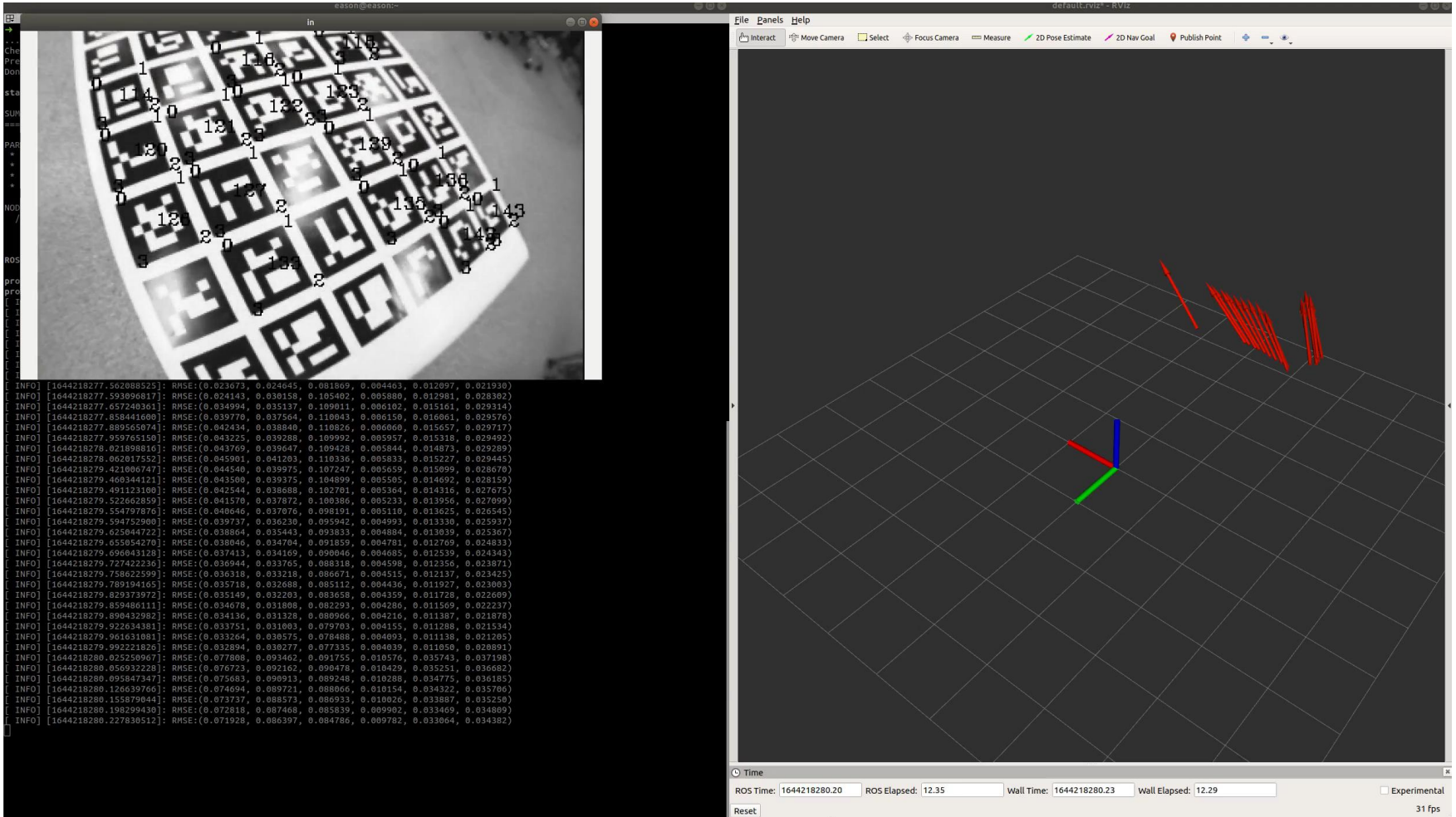
Project 2

- Phase 1: PnP-based localization on marker map
 - ROS & C++, Desktop PC, offline
- Phase 2: Visual odometry in markerless environment
 - ROS & C++, Desktop PC, offline

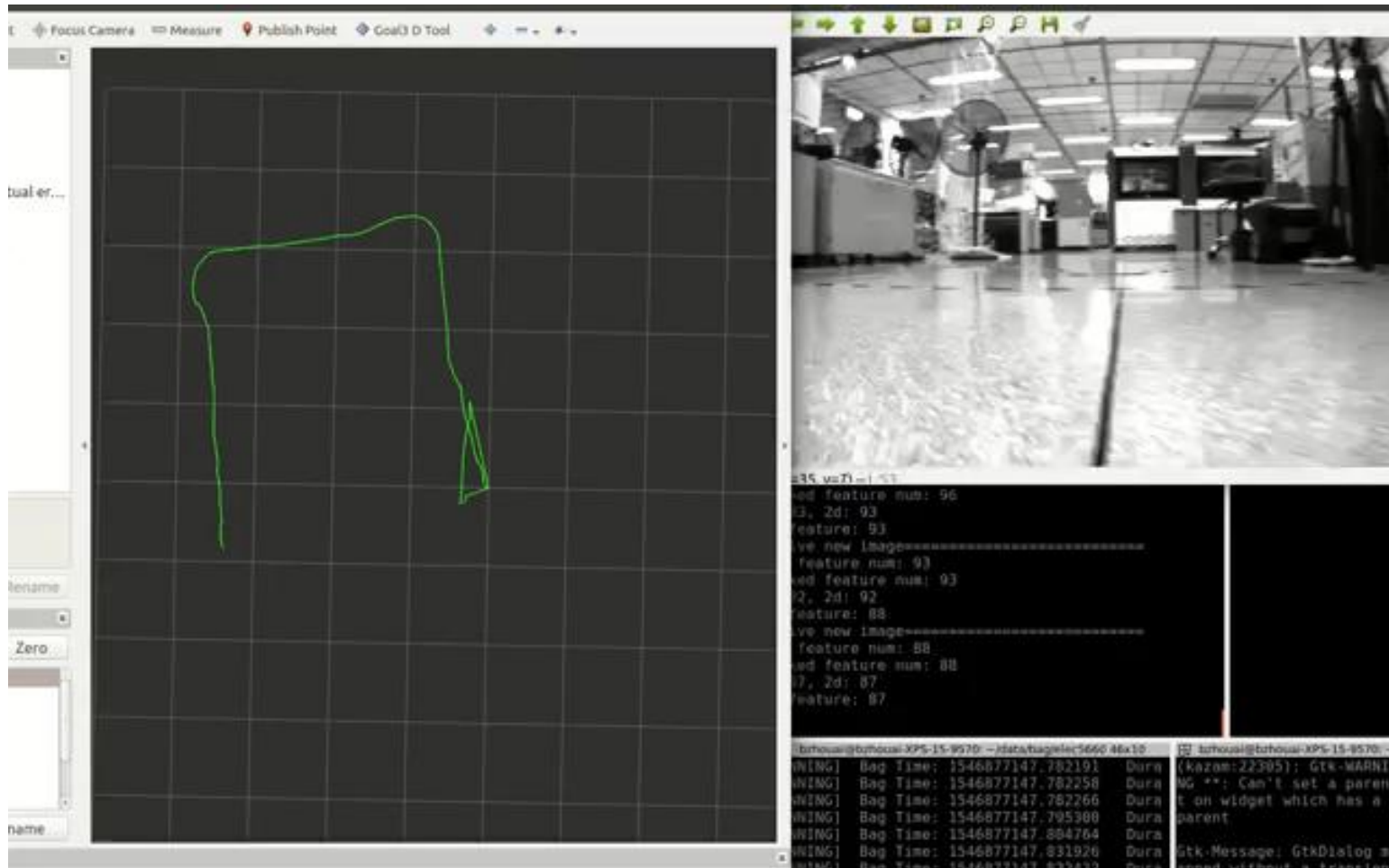


(b)

Project 2 Phase 1 in Past Years

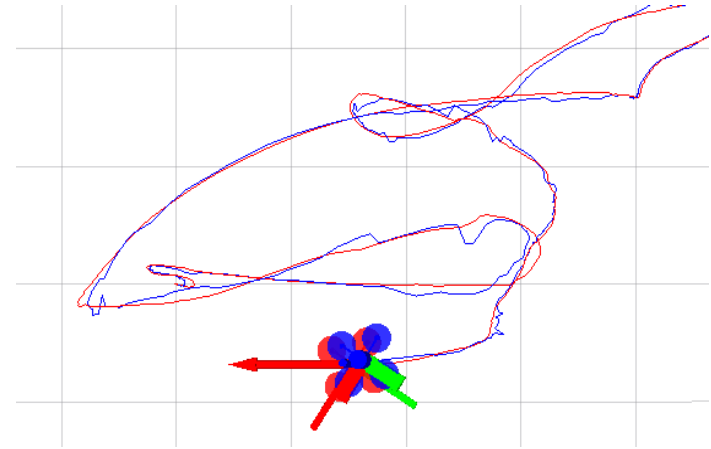


Project 2 Phase 2 in Past Years

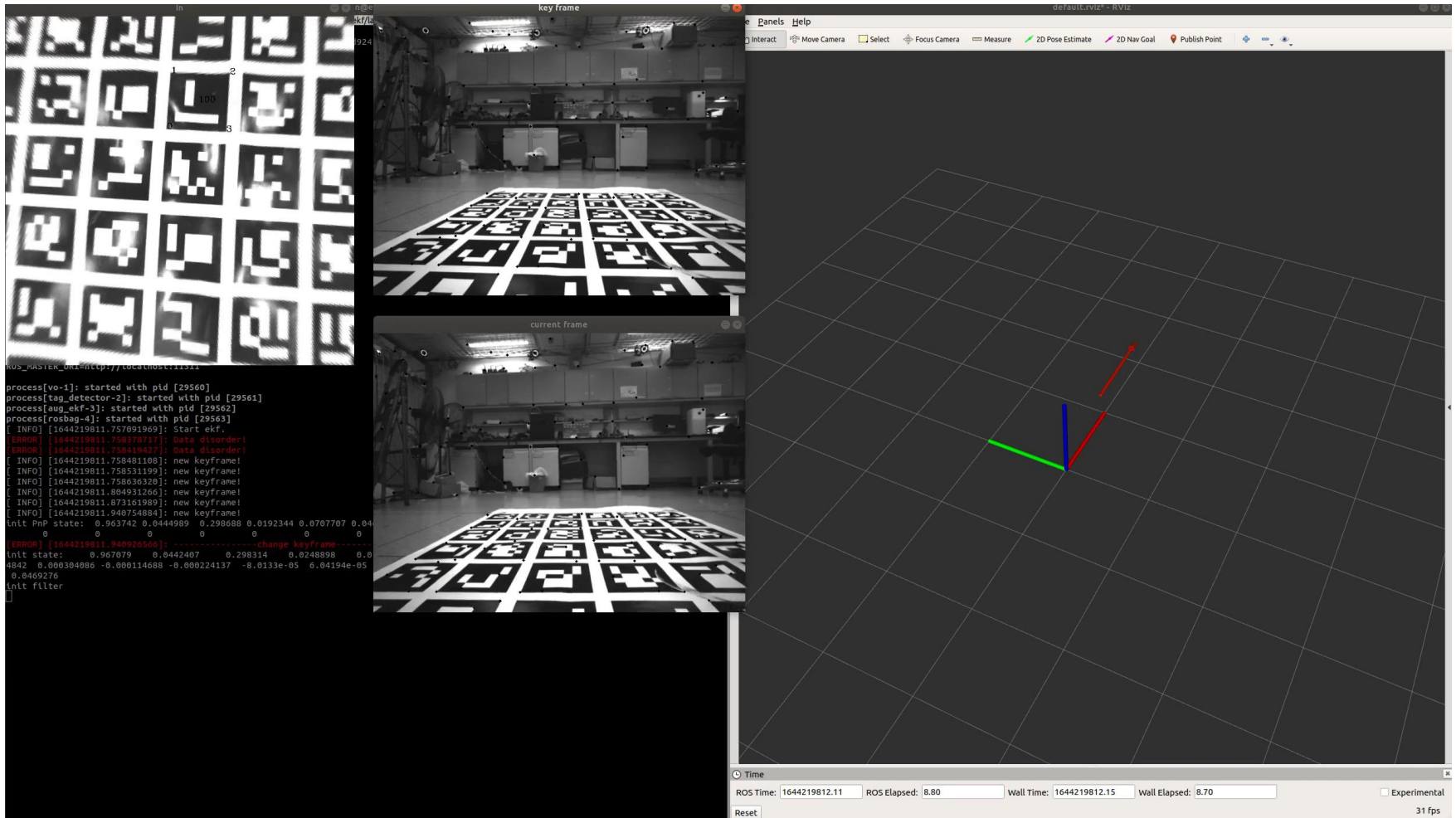


Project 3

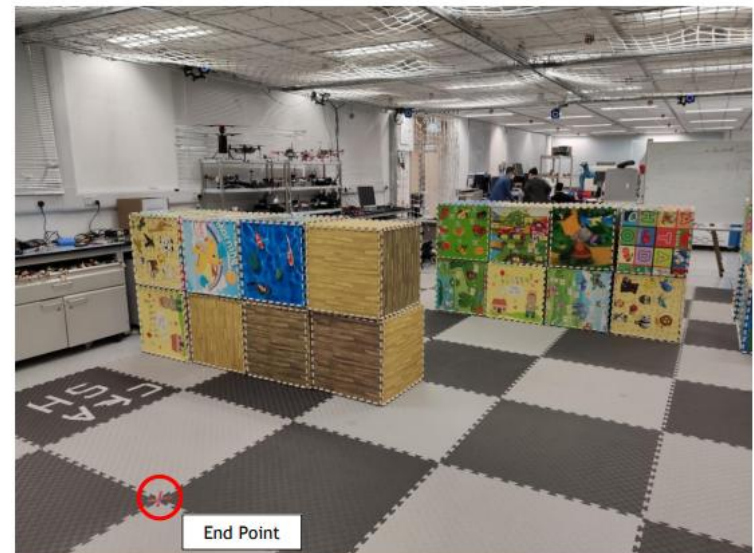
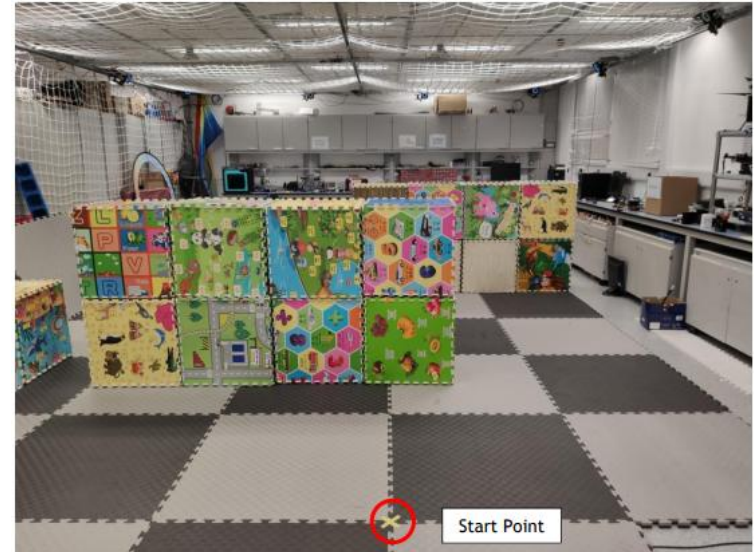
- Phase 1: EKF-based sensor fusion of IMU and tag-based position estimation
 - ROS & C++, Desktop PC, offline
- Phase 2: Augmented state EKF for fusion of IMU and keyframe-based visual odometry together with the PnP localization on markers
 - ROS & C++, Desktop PC, offline
- Phase 3: Fly the robot!



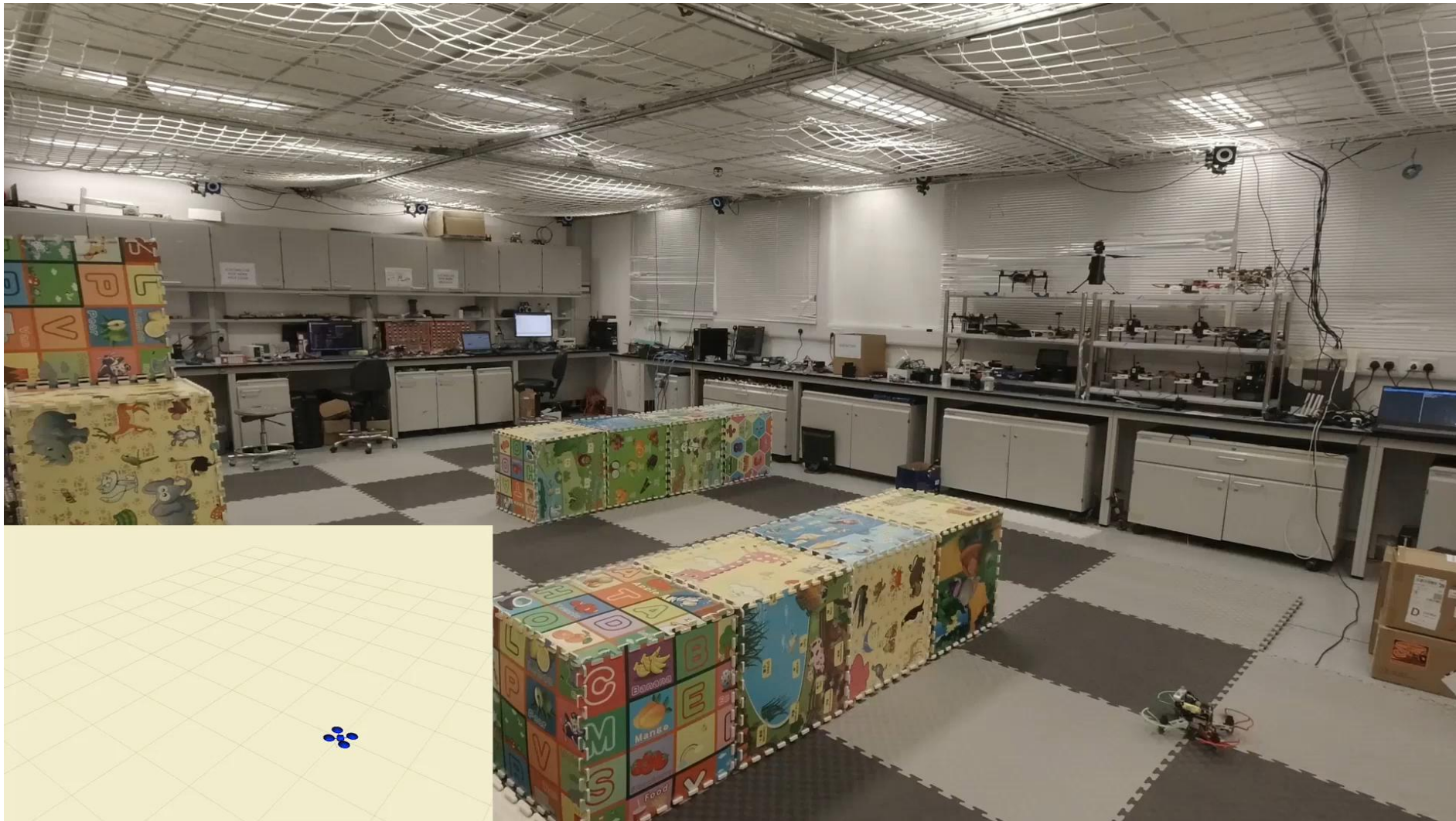
Project 3 in Past Years



Project 3 in Past Years



Project 3 in Past Years



Next Lecture...

- Rigid Body Transformations
- Rotational Motions
- Rotation Representations
- Rigid Body Motions
- Rigid Body Velocities
- Quadrotor Dynamics